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Magnetic disk device and method of controlling the same

Abstract

According to one embodiment, a magnetic disk device includes a first write control section of writing first data and second data to each of the sectors of a predetermined track among the tracks, a second write control section of writing the first data to each of the sectors of a predetermined track among the tracks, and a selection control section of, when receiving a write command, estimating the time required from start of a process of the first write control section to end of the process, and selectively executing the process of the first write control section or a process of the second write control section by comparing the estimated time with a time limit specified in advance.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-062233, filed Apr. 8, 2024, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to a magnetic disk device comprising a magnetic disk and a magnetic head, and a method of controlling the magnetic disk device.

BACKGROUND

(3) A magnetic disk device comprising a magnetic disk and a magnetic head executes writing and reading data to and from the magnetic disk in response to write and read commands transmitted from an external host device (host computer).

(4) When a so-called latency limit, which limits the delay in writing data, is specified by the host device, the magnetic disk device executes the data write process within the time limit of the latency limit.

(5) As data to be written to the magnetic disk, there are user data, which is the first data, and auxiliary second data such as error correction codes (ECC) for the user data and log data related to writing the user data. When the end timing of the time limit of the latency limit arrives during writing the second data, the magnetic disk device needs to stop writing the second data. In this case, the magnetic disk device suddenly needs to execute a procedure of forcibly writing the second data that could not be written, and the performance of subsequent data write is degraded due to the

sudden execution of this procedure. This performance degradation leads to delay in writing the data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram showing an overall configuration of each embodiment.

(2) FIG. **2** is a diagram showing a configuration of main portions of the magnetic disk in each embodiment.

(3) FIG. **3** is a diagram showing one track and plural sectors of a magnetic disk in the first embodiment.

(4) FIG. **4** is a diagram showing a data write range for each sector of FIG. **3**.

(5) FIG. **5** is a diagram showing a data read range for each sector of FIG. **4** and a situation in which data in unreadable sectors is rescued with an error correcting code.

(6) FIG. **6** is a diagram showing a time limit (ti+ta) of the latency limit in each embodiment.

(7) FIG. **7** is a diagram showing an example in which the end timing of the time limit ta arrives during the data write process in each embodiment.

(8) FIG. **8** is a flowchart showing control in each embodiment.

(9) FIG. **9** is a diagram showing a relationship between an expected time tx and the time limit ta in the first embodiment.

(10) FIG. **10** is a diagram showing one track and plural sectors of a magnetic disk in a second embodiment.

(11) FIG. **11** is a diagram showing a relationship between an expected time tx and a time limit ta in the second embodiment.

DETAILED DESCRIPTION

(12) In general, according to one embodiment, a magnetic disk device includes a circular magnetic disk including a plurality of tracks for data recording composed of a plurality of sectors arranged in a circumferential direction, the tracks being arranged in a radial direction, a magnetic head provided to seek in the radial direction of the magnetic disk and which writs and reading data to and from the magnetic disk, and a controller which controls rotation of the magnetic disk and seek of the magnetic head. The controller includes a first write control section of writing first data and second data to each of the sectors of a predetermined track among the tracks, a second write control section of writing the first data to each of the sectors of a predetermined track among the tracks, and a selection control section of, when receiving a write command, estimating the time required from start of a process of the first write control section to end of the process, and selectively executing the process of the first write control section or a process of the second write control section by comparing the estimated time with a time limit specified in advance.

(1) First Embodiment

(13) A first embodiment will be described hereinafter with reference to the accompanying drawings.

(14) As shown in FIG. **1**, a magnetic disk device **1** includes a magnetic disk **2** serving as a recording medium, a spindle motor **3** that drives rotation of the magnetic disk **2**, and a magnetic head **10** that writes and reads data to and from the magnetic disk **2**. An actuator **20** supporting the magnetic head **10** is arranged near the magnetic disk **2**.

(15) The actuator **20** supports the magnetic head **10** in a radial direction of the magnetic disk **2** so as to seek. In other words, the actuator **20** is also referred to as an actuator block or head stack assembly (HSA), and includes a rotary shaft **21**, an arm **22** with a middle part held by the rotary shaft **21**, a voice coil motor (VCM) **23** provided at a proximal portion of the arm **22**, a suspension member **24** provided at a distal portion of the arm **22** to hold the magnetic head **10**, and the like,

and rotates the magnetic head **10** from a first position P**1** represented by a broken line to a second position P**2** represented by a solid line by supplying a drive current to the voice coil motor **23**. In accordance with this rotation, the magnetic head **10** seeks (moves) in the radial direction of the magnetic disk **2** in a locus T shown in the drawing.

(16) A stopper ST and a ramp mechanism RL are arranged near the actuator **20**. The stopper ST limits the position movement of the magnetic head **16** on the inner circumferential side of the magnetic disk **12**. The ramp mechanism RL limits the position of movement of the magnetic head **16** on the outer circumferential side of the magnetic disk **12**.

(17) The magnetic disk device **1** includes a controller **30** that serves as the center of control, a head amplifier **41** that drives the magnetic head **10**, a signal processing circuit **42** provided in the connection between the head amplifier **41** and the controller **30**, a motor driver **43** provided in the connection between the voice coil motor **23** and the controller **30**, a DRAM **44** that is a memory storing programs, and the like necessary for the control of the controller **30**, a flash ROM **45** that is a memory storing various data necessary for the control of the controller **30**, a hard disk controller (HDC) **46** provided in the connection among the controller **30**, the hard disk controller (HDC), and an external host device (host computer) **50**, and the like.

(18) The head amplifier **41** amplifies data signals for write from the signal processing circuit **42** to the magnetic head **10** and also amplifies data signals to be read by the magnetic head **10**. The signal processing circuit **42** appropriately processes data signals for write from the controller **30** to the magnetic head **10** and supplies the signals to the head amplifier **41**, and also appropriately processes read data signals amplified by the head amplifier **41** and supplies the signals to the controller **30**. The motor driver **43** controls the drive current for the spindle motor **3** and the drive current for the voice coil motor **23** of the actuator **20** in response to instructions from the controller **30**.

(19) As shown in FIG. **2**, the magnetic disk **2** has a circular shape fitted coaxially with the rotational axis of the spindle motor **3** and includes a plurality of tracks Tr for recording data, which are formed to be arranged in a radial direction. As shown in FIG. **3**, each track Tr includes a plurality of first sectors S**1**, S**2**, . . . , Sn aligned in the circumferential direction of the magnetic disk **2** and at least one second sector Sp following the end of the arrangement of these first sectors S**1**, S**2**, . . . , Sn.

(20) The first sectors S**1**, S**2**, . . . Sn are sectors for recording, for example, the user data used by a user of the magnetic disk device **1**, which is main first data. The second sector Sp is a sector for recording, for example, track Error-Correcting Code (ECC), which is auxiliary second data related to the user data. The track ECC is an error-correcting code that executes error detection and correction of the user data for each track Tr.

(21) The controller **30** includes as main functions a position capture section **30***a*, a seek control section **30***b*, a settling determination section **30***c*, a first write control section **30***d*, a second write control section **30***e*, and a selection control section **30***f*.

(22) [Position Capture Section **30***a]*

(23) The position capture section **30***a* captures a position of the magnetic head **10** on the magnetic disk **2**, based on the position data of a servo pattern SB included in the read data of the magnetic head **10**.

(24) [Seek Control Section **30***b]*

(25) The seek control section **30***b* seeks the magnetic head **10** from a stop position on the magnetic disk **2** to a target position by controlling the drive (drive current) of the voice coil motor **23** of the actuator **20**. More specifically, the seek control section **30***b* seeks the magnetic head **10** from the stop position on the magnetic disk **2** to the target position, including acceleration and deceleration in order, by controlling the drive (drive current) of the voice coil motor **23** of the actuator **20**, based on the capture position of the position capture section **30***a*.

(26) [Settling Determination Section **30***c]*

(27) The settling determination section **30***c* executes so-called settling determination, which determines that the magnetic head **10** has reached the target position when a certain period has elapsed in a state in which the capture position Pos of the position capture section **30***a* is within a predetermined range including the target position.
(28) [First Write Control Section **30***d]*
(29) The first write control section **30***d* seeks the magnetic head **10** to a predetermined track Tr among the tracks Tr of the magnetic disk **2** under the control of the seek control section **30***b* and, writes user data (first data) to the predetermined number of first sectors from the beginning, among the first sectors S**1**, S**2**, . . . Sn on the predetermined track Tr in accordance with the rotation of the magnetic disk **2**, writes the track ECC (second data) to the second sector Sp of the above predetermined track Tr in accordance with the rotation of the magnetic disk **2** after writing the user data is completed, and notifies the host device **50**, which is the transmitter of the write command, of the completion of the process (status response code) after writing the track ECC is completed.
(30) That is, as shown in FIG. **4**, for example, user data is written to the three first sectors S**1**, S**2**, and S**3**, and after writing this user data is completed, the track ECC is written to the last second sector Sp as the magnetic disk **2** rotates (after waiting for a predetermined period of rotation). The combined period of the period of waiting for the rotation and the period of writing the track ECC to the second sector Sp is the track ECC period.
(31) For example, if the data in the first sector S**3** shown in FIG. **5** cannot be read when reading data, the error in the data in the first sector S**3** is detected and corrected by the track ECC in the second sector Sp by writing the track ECC. In other words, the data in the first sector S**3** is rescued by the track ECC.
(32) [Second Write Control Section **30***e]*
(33) The second write control section **30***e* seeks the magnetic head **10** to a predetermined track Tr among the tracks Tr of the magnetic disk **2** under the control of the seek control section **30***b* and, writes user data (first data) to the predetermined number of first sectors from the beginning, among the first sectors S**1**, S**2**, . . . Sn on the predetermined track Tr in accordance with the rotation of the magnetic disk **2**, and notifies the host device **50**, which is the transmitter of the write command, of the completion of the process (status response code) after writing the user data is completed.
(34) [Selection Control Section **30***f]*
(35) When receiving a write command from the external host device **50**, the selection control section **30***f* estimates time tx required from the start of processing (start of seek) to the end of processing (completion of transmitting the write command) in the first write control section **30***d*, compares the estimated time tx with time limit ta specified in advance, and selectively executes either the processing in the first write control section **30***d* or the processing in the second write control section **30***e*.
(36) More specifically, the selection control section **30***f* executes the processing of the first write control section **30***d* when the estimated time tx is the same as or shorter than the time limit ta or executes the processing of the second write control section **30***e* when the estimated time tx is longer than the time limit ta.
(37) The time limit ti is an element of the time limit (ti+ta) of the latency limit specified by 1 Command Duration Limits (CDL) function of the host device **50**, and is an inactive time limit that limits the write preparation time including predetermined preprocessing from the receipt of the write command to the time immediately before the disk access (immediately before the seek), as shown in FIG. **6**. The write preparation time is set to be within this time limit ti.
(38) The time limit ta is also an element of the time limit (ti+ta) of the latency limit specified by the CDL function of the host device **50**, and is an Active Time Limit that limits the write execution time including “seeking magnetic head **10**”, “writing user data”, “track ECC”, and “status response” from the start of disk access (start of seek) to the transmission of the status response code, as shown in FIG. **6**.

(39) A code specifying this time limit ti and ta is included in the write command transmitted from the host device **50** to the magnetic disk device **1**. By specifying the time limit ta, the host device **50** requires the magnetic disk device **1** to execute the data write process in response to one write command within the time limit ta.
(40) If the track ECC is to function in addition to the CDL function of the host device **50**, the end timing of the time limit ta of the latency limit may arrive during writing the track ECC to the last second sector Sp, as shown in FIG. **7**.
(41) If the end timing of the time limit ta of the latency limit arrives during writing the track ECC, the controller **30** needs to stop writing the track ECC. In this case, the magnetic disk device **1** suddenly needs to execute a procedure of forcibly writing the track ECC data that could not be written, and the performance of subsequent data write is degraded due to the sudden execution of this procedure. This performance degradation leads to delay in writing the data.
(42) The controller **30** executes the control shown in the flowchart in FIG. **8** to prevent this problem from occurring.
(43) When receiving a write command from the host device **50** (ST**1**), the controller **30** estimates the time tx required from the start of the processing (start of seek) to the end of the processing (completion of transmitting the write command) in the first write control section **30***d* (ST**2**).
(44) More specifically, the controller **30** obtains the write execution time including “seeking magnetic head **10**”, “writing user data”, “track ECC”, and “status response” from the start of the disk access (start of seek) to the transmission of the status response code, based on control data stored in advance in an internal memory or by calculation, and estimates the obtained write execution time as time tx required from the start of processing in the first write control section **30***d* to the end of processing (ST**2**).
(45) Then, the controller **30** compares this estimated time tx with the time limit ta specified by the CDL function of the host device **50** (ST**3**).
(46) If shorter “ta**1**” as shown in FIG. **9** is specified as the time limit ta by the host device **50** and the estimated time tx is the same as or shorter than the time limit ta (YES in ST**3**), then the controller **30** seeks the magnetic head **10** to a predetermined track Tr among the tracks Tr of the magnetic disk **2** (ST**4**), and writes the user data to a predetermined number of first sectors among the first sectors S**1**, S**2**, . . . , Sn in the above predetermined track Tr, from the beginning, in accordance with the rotation of the magnetic disk **2** (ST**5**). After completing writing the user data, the controller **30** waits for the magnetic disk **2** to rotate for a predetermined period when no user data is written (ST**6**), and writes track ECC (second data) to the second sector Sp at the timing when the position of the second sector Sp of the predetermined track Tr corresponds to the magnetic head **10** (ST**7**). After completing writing the track ECC, the controller **30** transmits a status response code indicative of the completion of the process to the host device **50**, which is the transmitter of the above write command (ST**8**).
(47) When receiving the status response code, the host device **50** executes a process of transmitting a next write command to the controller **30**.
(48) If longer “ta**2**” as shown in FIG. **9** is specified as the time limit ta by the host device **50** and the estimated time tx is longer than the time limit ta (NO in ST**3**), then the controller **30** seeks the magnetic head **10** to a predetermined track Tr among the tracks Tr of the magnetic disk **2** (ST**9**), and writes the user data to a predetermined number of first sectors among the first sectors S**1**, S**2**, . . . , Sn in the above predetermined track Tr, from the beginning, in accordance with the rotation of the magnetic disk **2** (ST**10**). After completing writing the user data, the controller **30** transmits a status response code indicative of the completion of the process to the host device **50**, which is the transmitter of the above write command (ST**8**).
(49) When receiving the status response code, the host device **50** executes a process of transmitting a next write command to the controller **30**.
(50) As described above, if the time tx from receipt of the write command to completion of

"writing user data", "writing track ECC", and "status response" is estimated and if the estimated time tx does not exceed the limit time ta of the latency limit specified by the CDL function of the host device **50**, the controller **30** executes a series of processes consisting of "writing user data", "writing track ECC", and "status response" to the end and can thereby write the track ECC to the second sector Sp without violating the latency limit.

(51) If the estimated time tx exceeds the limit time ta of the latency limit, only user data is written and the status response is made without writing the track ECC, thereby eliminating the inconvenience that writing the track ECC may be stopped during the writing.

(52) If no track ECC is written, error correction for the written user data cannot be executed. In consideration of this point, the controller **30** selects appropriate timing for preventing a subsequent data write performance from being degraded as write timing of the track ECC (i.e., track ECC in an unwritten state). In this case, since the controller **30** recognizes that writing the track ECC is not executed, not suddenly, but in advance, the controller **30** can accurately select appropriate timing that does not degrade the performance of writing the data, with sufficient margin. The controller **30** then writes the track ECC in an unwritten state to the magnetic disk **2** at the selected timing.

(53) Since the track ECC is written at the appropriate timing where data write performance is not degraded, the inconvenience that subsequent data write may be unnecessarily delayed can be avoided. The reliability of data recording can be thereby improved while avoiding violations of the latency limit.

(2) Second Embodiment

(54) A second embodiment will be described.

(55) As shown in FIG. **10**, each track Tr of a magnetic disk **2** includes a plurality of first sectors S**1**, S**2**, . . . , Sn aligned in the circumferential direction of the magnetic disk **2** and at least one second sector Slog following the end of the arrangement of these first sectors S**1**, S**2**, . . . , Sn. The first sectors S**1**, S**2**, . . . Sn are sectors for recording user data (first data) similarly to the first embodiment. The second sector Slog is a sector for recording auxiliary log data (second data) related to writing user data to the first sectors S**1**, S**2**, . . . Sn.

(56) Accordingly, a function of a first write control section **30***d* of a controller **30** is different from the function in the first embodiment.

(57) In accordance with a write command from a host device **50**, a first write control section **30***d* seeks the magnetic head **10** to a predetermined track Tr among the tracks Tr of the magnetic disk **2** under the control of a seek control section **30***b* and, writes user data (first data) to the predetermined number of first sectors from the beginning, among the first sectors S**1**, S**2**, . . . Sn on the predetermined track Tr in accordance with the rotation of the magnetic disk **2**, writes log data (second data) to the second sector Sp of the above predetermined track Tr in accordance with the rotation of the magnetic disk **2** after writing the user data is completed, and notifies the host device **50**, which is the transmitter of the write command, of the completion of the process (status response code) after writing the log data is completed.

(58) Processes in steps ST**2** and ST**3** in the flowchart shown in FIG. **8**, among the control executed by the controller **30**, are different from those in the first embodiment.

(59) The controller **30** obtains the write execution time including "seeking magnetic head **10**", "writing user data", "log data", and "status response" from the start of the disk access (start of seek) to the transmission of the status response code, based on control data stored in advance in an internal memory or by calculation, and estimates the obtained execution time as time tx required from the start of processing in the first write control section **30***d* to the end of processing (ST**2**). Then, the controller **30** compares this estimated time tx with the time limit ta specified by the CDL function of the host device **50** (ST**3**).

(60) If shorter "ta**1**" as shown in FIG. **11** is specified as the time limit ta by the host device **50** and the estimated time tx is the same as or shorter than the time limit ta (YES in ST**3**), then the controller **30** seeks the magnetic head **10** to a predetermined track Tr among the tracks Tr of the

magnetic disk **2** (ST**4**), and writes the user data to a predetermined number of first sectors among the first sectors S**1**, S**2**, . . . , Sn in the above predetermined track Tr, from the beginning, in accordance with the rotation of the magnetic disk **2** (ST**5**). After completing writing the user data, the controller **30** waits for the magnetic disk **2** to rotate for a predetermined period when no user data is written (ST**6**), and writes the log data to the second sector Sp at the timing when the position of the second sector Sp of the predetermined track Tr corresponds to the magnetic head **10** (ST**7**). After completing writing the track ECC, the controller **30** transmits a status response code indicative of the completion of the process to the host device **50**, which is the transmitter of the above write command (ST**8**).
(61) When receiving the status response code, the host device **50** executes a process of transmitting a next write command to the controller **30**.
(62) If longer “ta**2**” as shown in FIG. **11** is specified as the time limit ta by the host device **50** and the estimated time tx is longer than the time limit ta (NO in ST**3**), then the controller **30** seeks the magnetic head **10** to a predetermined track Tr among the tracks Tr of the magnetic disk **2** (ST**9**), and writes the user data to a predetermined number of first sectors among the first sectors S**1**, S**2**, . . . , Sn in the above predetermined track Tr, from the beginning, in accordance with the rotation of the magnetic disk **2** (ST**10**). After completing writing the user data, the controller **30** transmits a status response code indicative of the completion of the process to the host device **50**, which is the transmitter of the above write command (ST**8**).
(63) When receiving the status response code, the host device **50** executes a process of transmitting a next write command to the controller **30**.
(64) As described above, if the time tx from receipt of the write command to completion of “writing user data”, “writing log data”, and “status response” is estimated and if the estimated time tx does not exceed the limit time ta of the latency limit specified by the CDL function of the host device **50**, the controller **30** executes a series of processes consisting of “writing user data”, “writing log data”, and “status response” to the end and can thereby write the log data to the second sector Slog without violating the latency limit.
(65) If the estimated time tx exceeds the limit time ta of the latency limit, only user data is written and the status response is made without writing the log data, thereby eliminating the inconvenience that writing the log data may be stopped during the writing.
(66) If no log data is written, the log related to the written user data cannot be checked. In consideration of this point, the controller **30** selects appropriate timing for preventing a subsequent data write performance from being degraded as write timing of the log data (i.e., log data in an unwritten state). In this case, since the controller **30** recognizes that writing the track ECC is not executed, not suddenly, but in advance, the controller **30** can accurately select appropriate timing that does not degrade the performance of writing the data, with sufficient margin. The controller **30** then writes the log data in an unwritten state to the magnetic disk **2** at the selected timing.
(67) Since the log data is written at the appropriate timing where data write performance is not degraded, the inconvenience that subsequent data write may be unnecessarily delayed can be avoided. The reliability of data recording can be thereby improved while avoiding violations of the latency limit.
(68) The other constituent elements and control are the same as those of the first embodiment.
(69) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A magnetic disk device comprising: a circular magnetic disk including a plurality of tracks for data recording composed of a plurality of sectors arranged in a circumferential direction, the tracks being formed to be arranged in a radial direction; a magnetic head provided to seek in the radial direction of the magnetic disk and which writs and reads data to and from the magnetic disk; and a controller which controls rotation of the magnetic disk and seek of the magnetic head, wherein the controller includes: a first write control section of writing first data and second data to each of the sectors of a predetermined track among the tracks; a second write control section of writing the first data to each of the sectors of a predetermined track among the tracks; and a selection control section of, when receiving a write command, estimating the time required from start of a process of the first write control section to end of the process, and selectively executing the process of the first write control section or a process of the second write control section by comparing the estimated time with a time limit specified in advance.
2. The device according to claim 1, wherein when receiving the write command, the selection control section estimates the time required from the start of the process of the first write control section to the end of the process, and executes the process of the first write control section if the estimated time is the same as or shorter than the time limit specified in advance or executes the process of the second write control section if the estimated time is longer than the time limit specified in advance.
3. The device according to claim 1, wherein the magnetic disk includes the plurality of tracks for data recording composed of a plurality of sectors arranged in the circumferential direction and at least one second sector subsequent with the first sectors, and the tracks are formed to be arranged in a radial direction.
4. The device according to claim 3, wherein the first write control section writes the first data to each of the first sectors of the predetermined track among the tracks, and writes the second data to the second sector of the predetermined track, and the second write control section writes the first data to each of the first sectors of the predetermined track among the tracks.
5. The device according to claim 3, wherein the first write control section seeks the magnetic head to a predetermined track among the tracks, writes the first data to a predetermined number of first sectors from the beginning, among the first sectors of the predetermined track in accordance with the rotation of the magnetic disk, and writes the second data to the second sector of the predetermined track in accordance with the rotation of the magnetic disk after completing writing the first data, and the second write control section seeks the magnetic head to a predetermined track among the tracks, and writes the first data to a predetermined number of first sectors from the beginning, among the first sectors of the predetermined track in accordance with the rotation of the magnetic disk.
6. The device according to claim 1, wherein after completing writing the second data, the first write control section notifies a transmitter of the write command of completion of the process, and after completing writing the first data, the second write control section notifies the transmitter of the write command of completion of the process.
7. The device according to claim 1, wherein the time limit is a time limit of latency limit specified by a CDL function of an external device.
8. The device according to claim 1, wherein the first data is user data, and the second data is an error-correcting code executing error detection and correction of the user data for each of the tracks.
9. The device according to claim 1, wherein the first data is user data, and the second data is log data related to the user data.
10. A method of controlling a magnetic disk device comprising a circular magnetic disk including a

plurality of tracks for data recording composed of a plurality of sectors arranged in a circumferential direction, the tracks being arranged in a radial direction, a magnetic head provided to seek in the radial direction of the magnetic disk and writing and reading data to and from the magnetic disk, and a controller controlling rotation of the magnetic disk and seek of the magnetic head, the method comprising: first write control of writing first data and second data to each of the sectors of a predetermined track among the tracks; second write control of writing the first data to each of the sectors of a predetermined track among the tracks; and selection control of, when receiving a write command, estimating the time required from start of a process of the first write control to end of the process, and selectively executing the process of the first write control or a process of the second write control by comparing the estimated time with a time limit specified in advance.